Personal Project Report - Gripatronic, Creating a Wearable Robotic Glove

Krishneet Singh

Planning:

Product Goals

The product I've created is the *Gripatronic*. This project aims to learn and apply the skills required for the project and, eventually, create a working robotic glove that someone can wear. The intended product of my project is a wearable glove and arm piece that can aid in hand strength using servo motors. The product is designed to increase safety for workers or assist people with low dexterity. However, it is also intended to explore my passion for robotics further and learn more about every avenue of engineering. As for my final product, I created a proof of concept prototype. A proof of concept is a prototype of an idea to ensure it is viable and a reasonable idea to keep working on. I believe the most important thing I learned during this project was following the design cycle. The cycle was general guidelines for me and how I knew I was progressing. The first part of the cycle was to analyze and inquire about what project I would do and how to make it into a functional and realistic project. I looked at videos of robotics-related projects that piqued my interest, and I stumbled upon a video on Linkedin of a robotic hand that followed the movement of a human hand. I knew what I wanted to make but didn't know how to start. However, analyzing and reading about similar projects made it easier to start from somewhere. I created rough sketches and 3D models of what I envisioned and soon transferred my 2D designs into 3D. I used Fusion 360, the software we learned in design class in middle school; however, we didn't learn it to a great extent.





Learning Goal

This product will push me to explore many different and challenging topics I have no prior experience with. The field of robotics will be evolving at an alarming rate, and I believe it will be a better time than ever to experiment and explore robotics. The learning goal for this project will be to create a workable prototype based on an idea. Although very vague, I will want to experience

making something by myself and using my creativity. I want to know that I can create something with my two hands and that I won't be limited to what I learned. I have always told myself that something is too complicated and that I'm not qualified to do something, even if it's a good idea, but I will want to change that. I want to experience what it's like to be a designer. Within this project, I will need to learn how to code, specifically in C++, an essential skill for any automated project such as Gripatronic. I will also need to learn how to wire and solder, use an Arduino to use my code in my project, and learn about different sensors that can be used. In this case, I will use a rotary encoder to sense rotation and send signals to the Arduino.



```
void setup() {
  pinMode (outputA, INPUT);
  pinMode (outputB, INPUT);

  Serial.begin(9600);
  aLastState = digitalRead (outputA);

  myservo.attach(2);
}

void buttonReset () {
  buttonState = digitalRead(button);

  if (buttonState == HIGH) {
    counter = counter;
  }else {
    reverse ++;
  }
}



void encoderRead() {
  aState = digitalRead (outputA);
  if (aState != aLastState) {
    if (digitalRead (outputB) != aState) {
      counter ++;
    } else {
      counter --;
    }

    Serial.print("Position :");
    Serial.println(counter);
  }

  aLastState = aState;

}

void servoControl () {
  int degree = (counter * 9);
  if (degree >= userInput || reverse >= 1) {
    myservo.write(90);
    }
  else {
     myservo.write(180);
  }

}


void loop () {

  buttonReset();

  encoderRead();

  servoControl();
}
```

Additionally, for this project, I will need to learn how to create 3D models and use a 3D printer to create my models in the real world. In this project, 3D modeling and printing will play the most significant role in its success because I will need to constantly modify my project. All the technical components will be suitable; however, for every iteration and modification I make, I will have to either change the 3D model of the particular part or use different print settings. This project will make me uncomfortable and push me into situations that will be unfamiliar to me. Although it will be scary to do something I don't know anything about, I am passionate about robotics and am very excited to do this project.

Personal Interest

When asked what personal project I wanted to work on in the coming months, I immediately knew what to do. During the summer, I went to my friend's house to play a game called Call of Duty, where players could increase their strength and defense stats using an exoskeleton. An idea popped into my head immediately, as I wanted to create an exoskeleton. At first, it seemed like a far-fetched and nearly impossible product to make; however, I started to convince myself I could do it, and I also had a community with like-minded people all around me. I've been constantly surrounded by people who are designers, engineers, and hobbyists of robotics, myself included, and it was clear to me that I needed to pursue the idea of making an exoskeleton. I've had five years of experience with FIRST Robotics, which sparked my interest in robotics. It sparked my curiosity and allowed me to be a lot more creative in my daily life. I remember spending hours thinking about what was realistically possible with my current knowledge; however, I didn't like this approach. I realized I was cutting myself off with many potential

ideas because I was limiting myself to what I already knew instead of pushing myself to learn. One day I was on Linkedin, and I stumbled upon a video of a guy who created a robotic hand that tracked a person's hands and would move in correspondence to it. Videos like these were all over the internet, and each time I saw one, I'd get more motivated to create something of my own and go through the same process that all of these engineers and designers went through. I first tried to find what problem I was trying to solve, as it would allow me to implement this project outside of just a school activity. I wanted to somehow implement my love for strength training and athletics into robotics. I finally landed on creating a wearable device, similar to a powered exoskeleton, to increase a person's strength. I started developing various prototype sketches and saw which I liked best. I narrowed it down to the robotic glove as it was probably the project I didn't need a lot of materials for and the most intricate project of the sketches I made. The only problem was that I had no experience coding, wiring, soldering, or making a project like this. I had to learn and apply these skills to my project to succeed.

Success Criteria

When considering the success criteria for my robotic glove, I started by looking at what I wanted the product to do. I wanted to create a working glove that could open and close and have a stop feature. I made some guidelines and parameters for me to follow to achieve the intended purpose of my project. I wanted to push myself, so when creating the success criteria, I made goals more complex and challenging to reach but still achievable so I wouldn't lose confidence.

Category	Success Criteria	Description	Testing Strategy
Performance	1. The glove must be wearable	It is essential that users can put on and take off the glove because it isn't a permanent piece of technology. When putting the glove on or removing it, it shouldn't break or fall apart. I will do this by adding slits in my 3D prints to allow for the components of my design to flex and expand to the user's hand.	I will interview people and ask them to put on and take off the gloves and rate the ease of use from 1 to 10. Then, I will average the scores to get a rough estimate of how people feel about the wearability.
Human Factor	2. The product must be	I want to ensure the glove is comfortable and does	I will interview people, ask them to

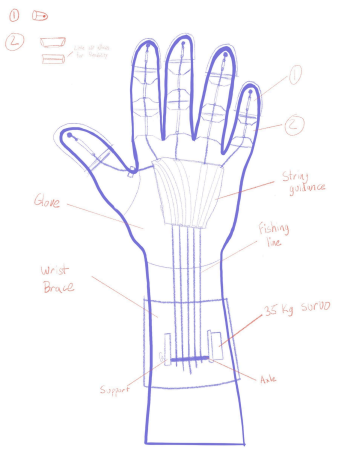
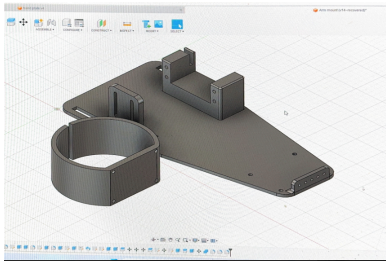
	comfortable for all users	not hinder people's movements. If the glove is uncomfortable, it would deter users from wearing it. I will make the glove comfortable by fileting (rounding) all sharp edges in the glove and using smooth PLA plastic for my 3D prints so it feels nicer on the user's skin. The glove will also be made of synthetic materials, reducing sweat.	put on the glove and rate its comfort from 1 to 10. Then, I will average the scores to estimate how people feel about the glove's comfort.
Human Factors	3. The product must be safe (Safety)	If I want to make this project, it must be safe. If the project isn't safe, it isn't viable, as it can harm the user rather than benefit them. I will create a button to override the code and stop the motor to make the glove safe. Additionally, I will use a weaker servo motor, which can allow the user to push against the servo motor if they must do so.	I will perform tests for the stop function in the hand. I will turn the servo on, then use the button to stop it and repeat it at least ten times to ensure the safety works.
Aesthetics	4. The product must be visually pleasing	For the product to be marketable or enticing to users, the product should fit a particular style and look. For this project, I am going with a black glove with golden parts, making it look very high-tech with wires, motors, and LEDs.	I will interview people and ask them to rate the aesthetics of the glove from 1-10. Then I will average the scores to estimate how people feel about the wearability
Performance	5. The product must be able to close and open	The glove's intended purpose is to aid the user in opening and closing	I will try to grab objects with the glove with different

	(Functionality)	their hand. If the product cannot fit this criteria, it isn't successful. I will measure this by letting my friends wear the glove and seeing if it helps them grip things.	textures and shapes to test if the glove can effectively close and open.
Personal	6. Understand the applications of this project (Knowledge)	To maximize my success in this project, I need to know what my product is, how it works, and how it connects to the world around us. I will research programming, mechanical building, and concepts such as exoskeletons and prosthetics.	

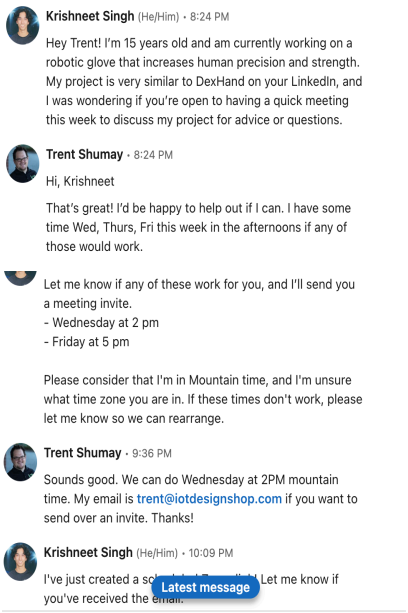
Action Plan

To create my action plan, I first brainstormed and compiled everything I wanted to do. Even if it wasn't achievable, I still wrote it down in my goals if I had the time to achieve them. Then I created a timeline resembling my action plan and created subgoals to accomplish the big ones. For example, when tasked with creating a 3D model of the components for the glove, I made mini goals to achieve them. Firstly, I had to buy the tools needed for this task: a digital caliper and download the software required to make the model. I then had to measure my hand dimensions to create the models. It required many mini steps to achieve the task, which is how I navigated through each other step of this process. Using these mini-tasks, I could keep myself on track to achieve bigger goals and track my progress in how I was doing with my given time. The Action Plan is a combination of this project's timeline and my goals for this project.

Status	Action	What was Done	Connection to Success Criteria	Comments	Deadline
Complete	Create a drawing and sketch of the design	I've created two drawings of my hand. One was on paper, my first brainstorming for the project, and the second was digitally so I could share my ideas with others.	Aesthetics	The drawings are complete and need to be converted into a 3D model. These drawings were made by looking at other projects, such	October 5th

				as the video from Trent Shumay.	
Complete	Creating 3D models with sketches	I used Fusion 360 to create the design's ring, cap, axle, and arm plate components. Using a vernier caliper, I could find the dimensions of all my parts and my hand measurements, which could be used in the 3D models. 	Performance	testing out different end caps and ring piece sizes.	Originally December 1st Finished: January 25th
Complete	Print parts, modifying if needed	I used a Prusa MK4 printer to print all my parts. I printed with PLA filament and changed the infill depending on how strong or flexible I wanted the pieces. I made the arm plate and axle sturdy, whereas the rings and caps were flexible. Modifications were needed with certain pieces, such as the encoder mount, as it wasn't strong enough and broke when placed under pressure.	Performance	Thumb: Large cap, large ring Index: Medium cap, 2 medium rings Middle: Large cap, 1 medium 1 large ring Ring: Large cap, 2 medium rings Pinky: small cap, 2 small rings	February 10th
Complete	Assemble prototype, test proof of concept	To assemble the mechanism, I attached one finger cap and two rings to the glove. I used the middle finger to check if my components were centered	Performance/ Human Factor	Was fully complete, 3d prints were glued onto the glove, and strings could move freely to pull the fingers in. .	February 25th

		and aligned. Then I fed the fishing line through with the holes in the pieces and tied the wire to the axle. The fishing line pulled the components, and my finger retracted, and the finger closed.			
Complete	Meeting with an advisor (Mr. Race)	We discussed how to make progress on my project. In our first meeting, we discussed my idea and whether the project was reachable and obtainable for the time I had. In the second meeting, we discussed my challenges, such as how to mount my Arduino board or manage wiring. In the 3rd meeting, we discussed some issues I was having with the coding, as well as some final touches to make the product look as clean as possible.	All	3/3	No deadline
Complete	Code Arduino	 Mentioned above	Performance/ Human Factor	Code the Arduino to stop the servo after several rotations and code the button as an emergency off switch.	January 1st (Completed December 16th)
Complete	Writing an article about exoskeletons	I decided to write a short article about exoskeletons to further enhance my knowledge about my product. I did a lot of research to understand the different mechanics of an exoskeleton and their applications in our lives. Additionally, I hope it allows me to contact new people by showing them that I am actually interested in robotics and aren't just saying things without anything to prove it. Hopefully, the article will act as something I	Personal	https://medium.com/@krishneet.gabri/breaking-human-limitations-e27354799e3a	No deadline

		can show off when I contact people I seek guidance from.			
	Connect with Experts	I practiced my communication skills online with people who I didn't know. I practiced a "cold approach" method to communicate and set up a meeting with an expert named Trent Shumay. Messages on Linkedin:  **Krishneet Singh** (He/Him) • 8:24 PM Hey Trent! I'm 15 years old and am currently working on a robotic glove that increases human precision and strength. My project is very similar to DexHand on your LinkedIn, and I was wondering if you're open to having a quick meeting this week to discuss my project for advice or questions. **Trent Shumay** • 8:24 PM Hi, Krishneet That's great! I'd be happy to help out if I can. I have some time Wed, Thurs, Fri this week in the afternoons if any of those would work. Let me know if any of these work for you, and I'll send you a meeting invite. - Wednesday at 2 pm - Friday at 5 pm Please consider that I'm in Mountain time, and I'm unsure what time zone you are in. If these times don't work, please let me know so we can rearrange. **Trent Shumay** • 9:36 PM Sounds good. We can do Wednesday at 2PM mountain time. My email is trent@iotdesignshop.com if you want to send over an invite. Thanks! **Krishneet Singh** (He/Him) • 10:09 PM I've just created a so... Let me know if you've received the email. Latest message	Personal	It was a precious experience, and he told me how his passion for robotics drove him to where he is today. He gave me a lot of ideas for my hand, including using a particular type of material so the strings wouldn't get stuck. He also loved the idea and has never seen a wearable robotic glove like the one I'm making. I will make sure to update him when I finish my project.	No deadline

Applying Skills

Achieving the Learning Goal - ATL Skill: Research

Research Question: What should I know to be knowledgeable enough to make a robotic glove?
Complementary Questions: Components: What do I need to automate a project? - What components are needed? What does an Arduino do? How to code an Arduino? - How to code in c++ How to use a breadboard? How to wire and solder? How to 3D print with different filaments and infills?

How strong are certain materials, and can they resist the forces I put on them?

Assembly:

What is the design process like?
How do I create accurate 3d models?
How do I create the perfect gap for pieces to interlock?
- How much slack do I need?

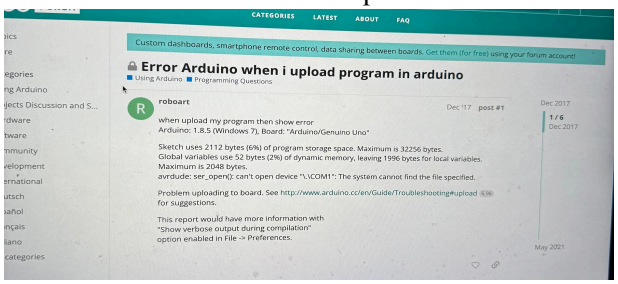
How do I accurately measure?
- How to effectively use a digital caliper

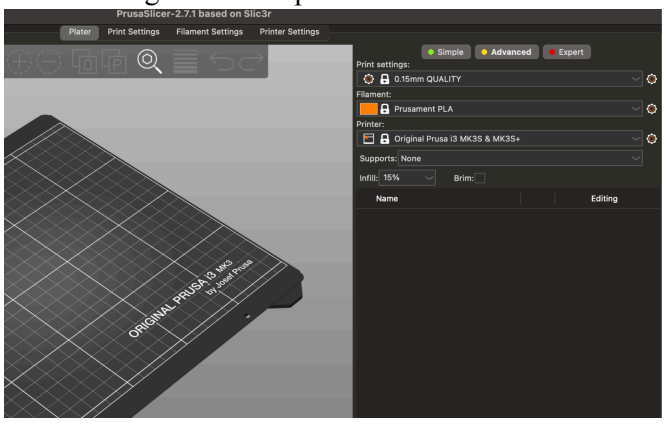
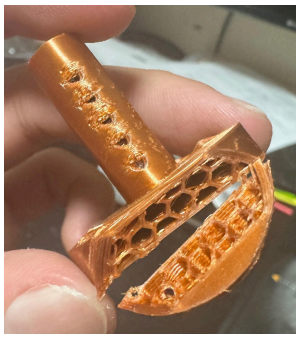
What are the steps to develop an idea into a workable product?

Sources	What did they do to aid me
https://www.linkedin.com/feed/update/urn:li:activity:7090397088391303168/?originTrackingId=%2FACHnGRAQzKWeIQX%2FmUQKw%3D%3D	This source has a video linked to it about a robotic technology called the DexHand. The hand uses motors and strings to control the fingers, and it served as motivation for my project. It helped me delve deep into what making something like the Gripatronic means and how I'd make it.
https://www.youtube.com/watch?v=OZ9g3O3X7dk	This video is about a robotic hand and the steps the creator went into making it. It was entertaining and gave me realistic hardships to overcome before starting the project, as well as motivation to start the project.
https://www.youtube.com/watch?v=ZzaPdXTrSb8	This video is a tutorial on how to use C++. C++ is the primary language of code used by my Arduino, and it was pivotal for me to know how to code it for my project to work and reach the success criteria.
https://www.youtube.com/watch?v=v4BbSzJ-hz4	This video explains how to use a rotary encoder and how to program it with the help of an Arduino.

ATL Skills Applied to Achieve Learning Goals

Making a robotic hand like *Gripatronic* involves many ATL skills. It requires many skills to create a successful product, and it is pivotal, especially in technology and engineering.

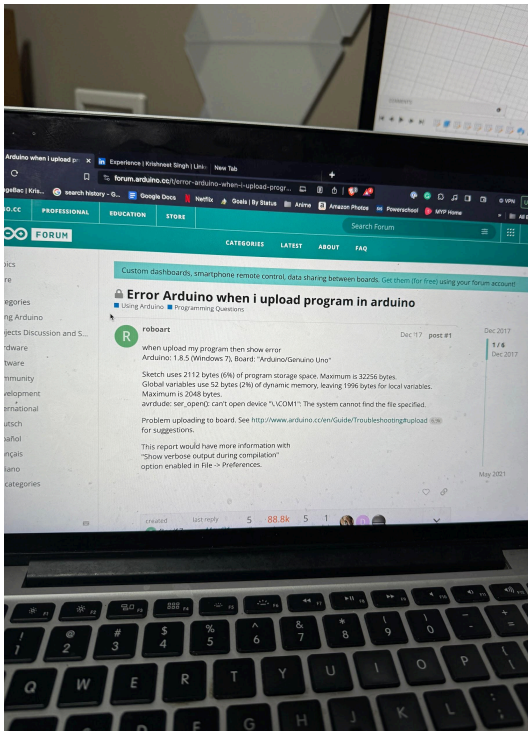
ATL skill	Description
Communication	I demonstrated communication by collaborating with experts such as Trent Shumay to discuss design ideas, share progress updates, and troubleshoot problems effectively. I received meaningful feedback on my project and took action to avoid future mistakes, such as using a stronger and frictionless string to make the hand smoother. I used various media to contact people, including email and Linkedin. These platforms allowed me to connect with people much more knowledgeable about the topics of my project that I was, and let me gain valuable insights about the struggles I was having and the future struggles I'd have in the future and how to avoid them.
Researching	I demonstrated practical research skills by researching existing robotic hand designs, materials, and technologies to understand how they work and what they are. I use various resources such as websites, articles, and videos to aid my research. I learned how to collect, record and verify data and deceiver whether the information I saw online was accurate. I reduced the amount of misinformation by researching trusted sources, such as the official Arduino Forum. I had to make connections between various sources to determine if what I was doing was right, and if they did differ, why? This process helped me debug my coding and find what was wrong with my Arduino, as I could refer back to multiple sources and combine the information to help me. 
Critical Thinking	I demonstrated critical thinking by analyzing the functionality of different robotic hand components and evaluating their strengths and weaknesses. I had to identify potential challenges in the design process and develop innovative solutions to overcome them. For example, one of the biggest problems I faced when thinking about the robotic hand was how the thumb would close. When your hand forms a fist, the thumb travels in a different direction from the other fingers, which is a huge pain. I had to use critical thinking to redirect the force of the string by using hinges and work my way around the problem. Before taking any steps for my project, I had to think critically about the benefits or risks of my actions. It led me to question the actual function of certain parts and reduce the work I had to do because the components were simply unnecessary. It taught me how to perceive risk management and allowed me to understand why I was doing things.
Problem-Solving	I demonstrated problem-solving skills by approaching challenges as smaller, manageable tasks instead of larger unreasonable ones. By brainstorming and contacting experts, I could generate multiple solutions and select the most effective one. Additionally, I had to learn how to adapt to unexpected obstacles,

	which made me more robust in the areas I was weak in. For example, when printing the axle for my project, the axle broke under the pressure of the strings and motor. To negate this problem, I had to learn how to increase the infill of a 3D print and make the prints denser. This not only taught me how to use software I didn't know a lot about but also taught me how it was okay to make mistakes because it let me reach closer to perfection. This is the PrusaSlicer, which is used to edit settings on the 3d printer.  

Reflecting

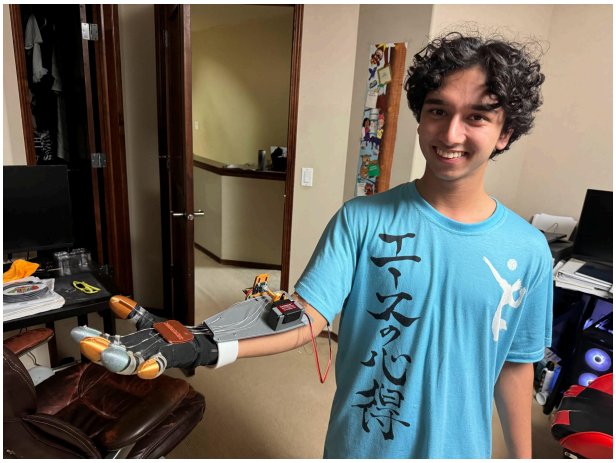
<u>Impact of the Project</u>

This project has impacted not only myself but hopefully has an impact globally. When starting this project, I'd never fully committed to creating something of my own, using my abilities and dedication. This project has completely changed how I view problems and think of myself. Before the project, I never ventured out of my comfort zone and only did the things I knew how to do. I'd never even dare to take on a project if it wasn't forced on me or because I felt like it. But after failing, experimenting, and eventually finding my way out of a deep hole, I completed something I am incredibly proud of. Creating something with my own two hands, with little to no knowledge about what I was doing, made me extremely happy about myself and my ability to create something from nothing. I feel like I can take and pursue any problem, project or idea because I now know how to battle against obstacles instead of avoiding them; I now have the experience to challenge myself. I have improved my physical abilities to create and engineer and have a completely different mindset when making ideas a reality. I've also gained so many valuable skills along the way. The first and most important skill I learned was figuring something out without relying on others. For the coding parts of my project, I didn't just use AI to write it



for me, but actually did the research and put the work in to figure it out. I even ran into multiple problems that frustrated me, but it was highly rewarding when I figured it out because I didn't rely on outside sources like my teachers or family. However, this project may stretch to a bigger audience than just myself. I believe this project will be highly influential in the near future, as our world is rapidly changing with automated technology. Humans are much weaker in many aspects of physical ability when compared to robots, and to enhance people, my product can help. I want to continue building on my project and hopefully make a product that maximizes a person's grip strength rather than just allowing it to close a person's hand like it is now.

Evaluation of Product





Success Criteria	Testing Results	Obtainability
The glove is wearable (Performance)	The glove is wearable for my hands. I can slide my hands in the glove, and the pieces expand due to the slits in them. However, for people with bigger hands, the glove won't fit as well and will feel tight. I asked ten people about the wearability of the glove, 8 of those people found it easy to wear, and two found it challenging. They gave it an average rating of 4 out of 5. SEMI COMPLETE	This goal was challenging yet attainable within the given amount of time. I just need more time to figure out how to make it fit not just me but every user.
The product is comfortable for all users (Human Factor)	When I first put it on, the glove felt relaxed; however, after lots of use, it got very sweaty. Additionally, the hard plastic of	This goal was easy but still necessary as a success criterion. Using comfy materials and round edges on models requires

	the arm plate doesn't feel natural and doesn't sit very well on my skin. Ten people were asked to rate the glove's comfortability. Six people said it was comfortable, and four people said it wasn't. It received a score of 3.5 out of 10 for comfortability. SEMI COMPLETE	little effort; however, if I had more time, I could make the glove much more comfortable. I could do this by curving pieces to fit over my arm instead of sitting on them or using thinner materials to reduce sweat.
The product is safe (Human Factor)	I programmed a button using the rotary encoder to stop the motor whenever it is pressed. This allows for an immediate stop when the user needs it. Additionally, I switched to a 25kg servo motor instead of a 35 to reduce the damage if something terrible happened and a person could resist it. COMPLETE	This was a challenging and attainable goal, and also necessary. I didn't want a dangerous product, so this was the first area I wanted to complete. This was challenging because I needed to rewire things to make the button work and create new code to tell the servo to stop after the button was clicked.
The product is visually pleasing (Aesthetics)	The product looks super cool with a high-tech and colorful vibe. However, each component has different colors; I will change this when I have time. Of the five people I asked, all five said it looked very cool, receiving a perfect score of 5 out of 5 for aesthetics. SEMI COMPLETE	This success criteria isn't good in my opinion, as it's very subjective and not accurate. I didn't know what to achieve, but I wanted the product to look cool. In my opinion, my glove seems pretty cool, so it does complete the success criteria.
The product must be able to close and open (Performance)	The product opens and closes sometimes because the wires get stuck on the axle and get tangled together. There is also no reverse function yet; the string must be manually unwrapped to close it again. NOT COMPLETE	This goal was not obtainable with the time I had. I expected to have about one hour a week to finish this project; however, it took much longer than that. Because of that, I couldn't completely meet this success criterion, and I had to settle with just closing my hands. However, I will still keep working on it.
Understand the applications of this project (Personal)	I wrote an article about my knowledge of exoskeletons and robotics from my research. COMPLETE	This was an obtainable goal that also aided me tremendously. I learned more about my project and the struggles other people were facing, how to negate

		them, and how to do effective research online.

Assessment of the Product

The key ATL skills I had to use were communication, researching, critical thinking and problem-solving skills.

Firstly, communication allowed me to have access to knowledge that I simply couldn't obtain during the time frame I had. This is because I could contact experts in the field, ask questions and ask for help when needed. It was an essential skill that significantly improved the quality of my product and the effectiveness of how I worked. This helped me fulfill my learning goal by helping me obtain the skills and insight into problems.

Secondly, the research skills I developed were pivotal in my success throughout this project. It allowed me to move past relying on others to get me out of situations that made me uncomfortable and persistent in finding an answer. When I encountered a problem, the first thing I would do was research everything about it, so I could effectively negate the issue and have enough information to avoid it if I reencounter it. It built resilience in me to keep at it when I didn't know why my code wasn't working, or the 3D prints weren't accurate enough.

Thirdly, critical thinking and problem-solving skills were the most essential skills for this project. When things weren't working, or fitting, or broke, my first instinct was to just give up, but I knew that if I did, I'd never make strides forward to complete my project. I learned to think critically about every tiny detail and use those details to find solutions around them. For example, when designing the 3D models for the parts, the screw holes were too narrow for the screws I wanted to fit. Usually, I could just increase the hole size, but in this case, increasing the size would allow the parts to move. I had to think critically and problem-solve about how to fix this problem, and I learned how to thread a hole in my models, which allowed me to use the same size hole and still fit the screw in.

These ATL skills tremendously helped me achieve my learning goal: making a working prototype based on my creativity. I couldn't have been successful in achieving my learning goal if I had not put in the dedication and time required, and because of the effort I put in, I learned and applied these ATL skills.

Some of the factors within control were the time I put into this project and the ideas. I could choose appropriate materials to ensure comfort and safety in the glove's design. I also had design freedom and could constantly test and prototype to see which parts fit best. Additionally, I could achieve my success criteria with the freedom I had. For example, I could create a comfortable, sweat-reducing glove that met safety standards using smooth PLA plastic and synthetic materials. However, one of the factors I couldn't

control was the supplier delays and mechanical issues. The delays in receiving components from suppliers hindered my project timeline and led me to follow a much more stressful and compact schedule. Many of the hardware components, such as the fried servo, broke, and I had to buy more parts. It halted progress because I couldn't test my code or the functionality of the glove without the components.

If I redo this project, I would prioritize better time management strategies to mitigate the impact of unforeseen delays and ensure a more successful completion. By improving my time management, I would enhance the project efficiency and reduce the likelihood of setbacks caused by factors I didn't have a choice in. I can't control what is happening in the world. However, I can account for events like shipping delays or parts breaking. This project spans far beyond just an assignment given to me by the school, but is something I want to continue to pursue. Although I will continue to make improvements, it is helpful to know the challenges and difficulties of making the *Generation 1* model so that when I make future prototypes, I can include these delays and uncertainties in my timeline.